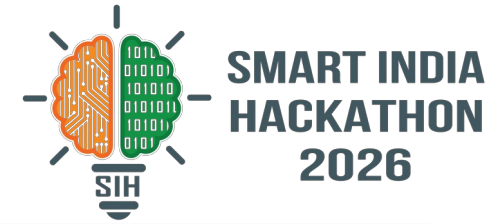


SMART INDIA HACKATHON 2026



1 PROBLEM STATEMENT MDoNER



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

PROBLEM STATEMENT ID
Smart Automation • Software

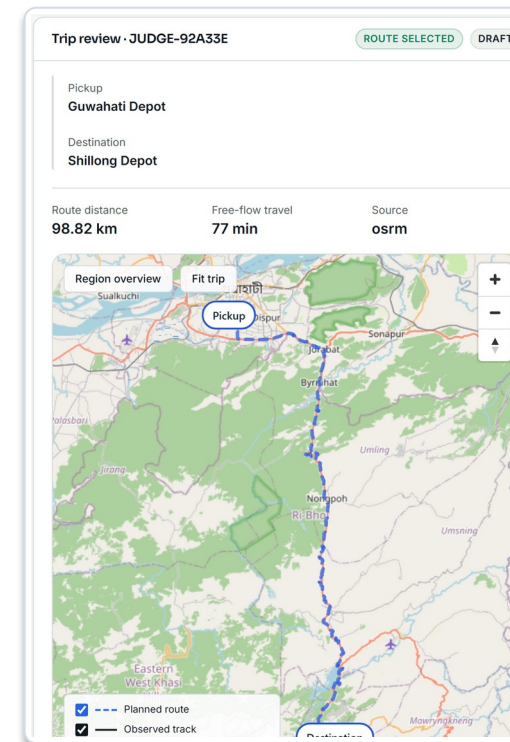
PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

SHORTEST IS NOT ALWAYS USABLE.

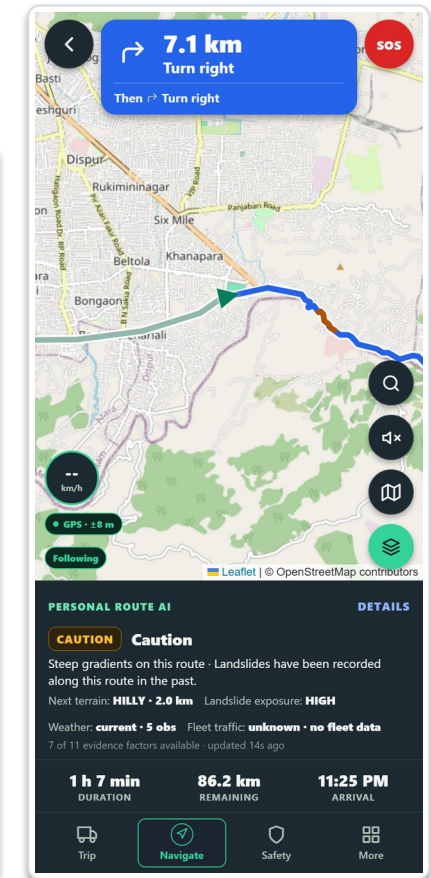
“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype • physical-phone validated • live demo corridor Guwahati → Shillong

2 WORKING SYSTEM manager console + Android driver app



Real screens • real road • no mock-ups



RASTA AI — ROUTE INTELLIGENCE

1 THE PROBLEM

Monsoon · landslides · floods · official warnings · single-lane hill roads · no signal.

A shortest-path router still sends the truck.

98.8 km	real road on the demo corridor	22	landslides within 5 km
1975 m	climb, steepest 10.3 %	154	NER events, NASA GLC

2 SHORTEST vs USABLE



3 WHAT MAKES IT DIFFERENT USP

CORRIDOR USABILITY, NOT SHORTEST PATH

11 factors scored on the selected road before dispatch

HUMAN-GOVERNED REROUTE

a real off-route triggers a real alternative; a manager accepts before the phone follows

UNKNOWN IS NEVER SAFE

missing or stale evidence blocks selection until a reviewer authorises

ONE APK, 23 LANGUAGES, OFFLINE

server-decided role; whole-trip package cached on the phone

4 EVIDENCE SCREEN

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN

Manager · Check conditions

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN — never safe — and goes to an authorised reviewer.

TECHNICAL APPROACH

1 INPUT verified sources, each with health + freshness

WEATHER

Open-Meteo · MET Norway

TERRAIN

Copernicus DEM ·
OpenTopoData

WARNINGS

NDMA SACHET (CAP)

FLOOD

GloFAS

LANDSLIDE HISTORY

NASA GLC 2007-17

TRAFFIC

fleet GPS fixes

2 PROCESS deterministic, auditable, no model in the loop

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · freshness per factor

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

3 OUTPUT · BUILT WITH seven technologies, nothing decorative

Re-scored every 60 s on the road ahead. A reroute is proposed to the manager, never applied silently.

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

CLIENTS Manager console (React + TS) · Driver app (React Native / Expo, Android) · one login, server-decided role, offline package

4 WHO DECIDES

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence

Deterministic safety policy

Manager governance

Driver action

FEASIBILITY AND VIABILITY

1 FEASIBILITY OF THE IDEA

TECHNOLOGY

FastAPI + PostGIS, two clients on one REST API. Every part is built and running.

IMPLEMENTATION

Hosted backend, manager console, Android APK 1.0.18. Certified on a physical phone.

COST

No licensed map, traffic or weather feed — every source is public or open.

SCALABILITY

A new source is a new provider adapter behind the same policy. Fleet traffic improves per truck.

2 RISKS → WHAT WE DO ABOUT THEM

Provider rate limit or outage

→ Cached evidence; the factor shows UNKNOWN, never assumed safe

Cold start on free hosting

→ Warm /health before the slot; state is server-side

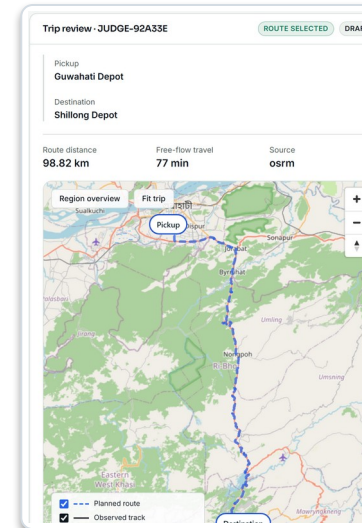
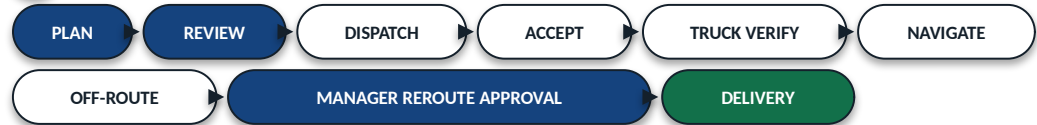
No phone signal on the corridor

→ Whole-trip offline package; navigation continues, labelled OFFLINE

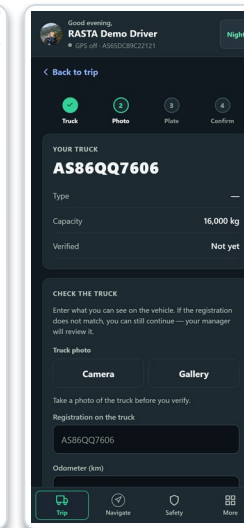
Experimental model unreliable

→ Safety gate holds it outside routing until it passes

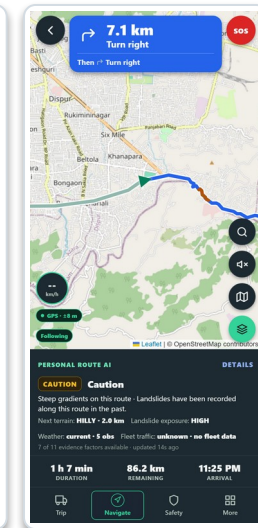
3 PROVEN, NOT PROPOSED physical phone · hosted backend · real road



Manager · review



Driver · truck check



Driver · navigation

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · **621+** driver · **170+** manager tests

Canonical demo Guwahati → Shillong ≈ 98.8 km real route

IMPACT AND BENEFITS

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

1 WHO BENEFITS

FLEET MANAGERS

Usability and uncertainty visible before dispatch. Auditable.

DRIVERS

Danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

Resilient access to food, medicine, fuel, relief.

GOVERNMENT / OPS

Auditable evidence and disruption per corridor.

2 MEASURABLE TODAY counted, not estimated

23 selectable languages, explicit fallback status

12/12 judge-flow steps certified on a physical phone

11 route factors in every decision

100 % of reroutes reviewed by a human

60 s route-ahead monitor while driving

0 unknown factors counted as safe

3 HUMAN GOVERNANCE AI has no uncontrolled authority

Verified evidence

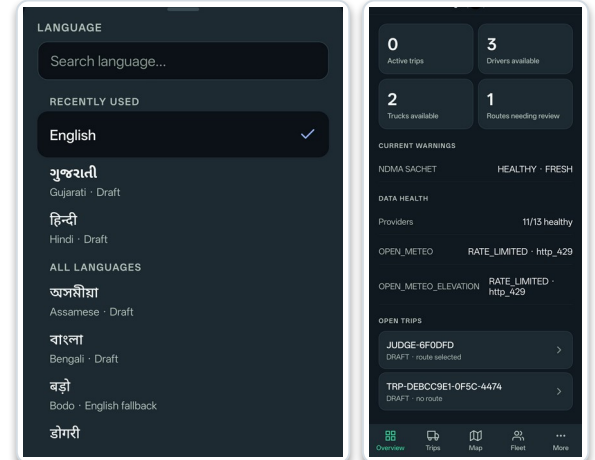
Deterministic safety policy

Manager governance

Driver action

Every reroute is proposed, reviewed and accepted by a person. No invented percentages: impact is stated as who gains what.

4 REAL SCREENS APK 1.0.18



23 languages

Manager • mobile

APK 1.0.18. One login, server-decided role. Every language shows its status — Verified, Draft or English fallback.

1 SOURCES BEHIND EVERY ROUTE DECISION

ROUTING / PLACES

OSRM • **OpenStreetMap** • **Nominatim**

Road geometry, alternatives, turn steps.

WEATHER / TERRAIN

Open-Meteo • **MET Norway** • **Copernicus DEM** • **OpenTopoData**

Forecast along the road; climb, gradient, steep segments.

OFFICIAL / FLOOD

NDMA SACHET (CAP) • **Copernicus GloFAS**

Official warnings with freshness; discharge vs 30-day mean.

LANDSLIDE HISTORY

NASA Global Landslide Catalog

Historical exposure, 2007–2017, ≤ 5 km accuracy.

AI ASSISTANCE

Google Gemini

Explains in the driver's language. Never safety authority.

PROBLEM STATEMENT

MDoNER • **SIH26002**

Smart logistics and accessibility intelligence for the NER.

2 EXPERIMENTAL ML research only

95 %

RECALL

NER geographic holdout • 154 events • no NER row in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment



VALIDATION BEFORE AUTOMATION. Recall was not enough. Held-out testing exposed the false-positive rate. The gate keeps it out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 — calibrated logistic regression, NASA GLC + ERA5-Land rain + DEM slope